

BANARAS HINDU UNIVERSITY

B.A. (Hons.) Semester V Examination, 2023-24

Paper No. STA-401: Sample Surveys and Design of Experiments

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

FRG 90 soveassejsoxavsonceessaestoncegemnitioents Statistics STA-401 : Sample Surveys and Design of Experiments Note: Answer ANY FIVE questions. All questions carry equal marks **Question 1.** What do you mean by a random sample? Discuss the simple random sampling procedure with its types and their properties. Show that the sample proportion is an unbiased estimator of the population proportion in both SRSWR and SRSWOR cases. Derive the expression of asymptotic confidence intervals of the population mean in the case of SRSWOR. [#1]

Question 2. What exactly do you mean when you say sample survey? Write down the advantages of a sample survey over a complete enumeration. What are the basic principles of a sample survey? [#1]

Question 3.

- (a) Show that $E(s^2) = \sigma^2$ for SRSWR and $E(s^2) = S^2$ for SRSWOR, where σ^2 denotes the population variance. [#1]
- (b) Discuss the methods of sample size determination using a pre-specified margin of error and standard error of an estimator. [#1]

Question 4. A population consists of four units with respective values 20, 34, 46, and 60. Write down all possible samples of size 2 using SRSWR and SRSWOR sampling schemes. Using selected samples, obtain the unbiased estimate of the population mean with standard errors for both cases. For the given data, verify that $E(s^2) = \sigma^2$ for SRSWR, $E(s^2) = S^2$ for SRSWOR, and $E(\bar{y}) = \bar{Y}$ for both cases. [#1]

Question 5. What is analysis of variance (ANOVA)? Write down its assumptions and two applications in real-life scenarios. Discuss ANOVA for one-way classified data in detail. [#1]

Question 6. Discuss stratified random sampling with its advantages. Is the sample mean an unbiased estimator of the population mean? If not, derive the unbiased estimator of the population mean with its standard error. Derive the expression of asymptotic confidence

intervals of the population mean.

[#1]

Question 7. What is design of experiments? Write down the basic principles of design of experiments. Discuss the randomized block design with its merits and demerits. [1]

Question 8.

- (a) How do you select samples using systematic sampling and cluster sampling procedures? Are these designs better than SRSWOR? State the unbiased estimator of the population mean for both sampling designs. [1]
- (b) Explain how you utilize auxiliary information to improve the estimation of population parameters. Discuss ratio and regression estimation for the population mean. [1]